



FERRIOXALATE-MEDIATED PHOTODEGRADATION OF ORGANIC POLLUTANTS IN CONTAMINATED WATER

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Abstract—The UV-visible photolysis of ferrioxalate in the presence of hydrogen peroxide (the UV-vis/ferrioxalate/H₂O₂ process) is investigated for the treatment of contaminated groundwater and industrial wastewater. This process generates the hydroxyl radical, which is a strong oxidant and reacts rapidly with most organic compounds present in solution. Tap water spiked with various organic pollutants, a process water containing chlorobenzene, a tank-bottom water containing a mixture of benzene, toluene and xylenes (BTX), a wastewater containing 1,4-dioxane and another wastewater containing methanol, formaldehyde and formic acid were treated with the UV-vis/ferrioxalate/H₂O₂ process and the efficiency compared with alternative oxidation processes, including the UV/H₂O₂ and UV-vis/Fe(II)/H₂O₂ (UV-Fenton) processes. The destruction of specific compounds and group parameters, such as TOC and BOD, were used to examine the relative energy efficiencies of the alternative processes. The data indicate that in nearly all cases, the new UV-vis/ferrioxalate/H₂O₂ process has a much higher efficiency (by a factor of about 3 to 30) than does either the UV-vis/Fe(II)/H₂O₂ process or the UV/H₂O₂ process. It is concluded that this new process is very efficient and useful for the treatment of moderate to highly contaminated waters. © 1997 Elsevier Science Ltd. All rights reserved

Key words—ferrioxalate, Fenton's reagent, photooxidation, hydroxyl radical, organic pollutants, photomineralization, wastewater

INTRODUCTION

The destruction and mineralization of organic pollutants in waste streams is now possible using Advanced Oxidation Processes (AOPs), which almost all rely on the generation of very reactive free radicals, such as the hydroxyl radical ($\cdot\text{OH}$). When generated, these radicals react rapidly and (usually) indiscriminately with most organic compounds, either by addition to a double bond or by abstraction of a hydrogen atom from aliphatic organic molecules (Buxton *et al.*, 1988). The resulting organic radicals then react with oxygen to initiate a series of degradative oxidation reactions that ultimately lead to mineralization products, such as CO₂ and H₂O (for a review of UV-based AOP systems in homogeneous solution, see Bolton and Cater (1994) and Legrini *et al.* (1993)).

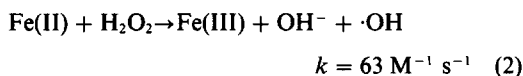
Currently, many commercial AOP installations utilize the UV photolysis of H₂O₂ to produce $\cdot\text{OH}$ radicals†.



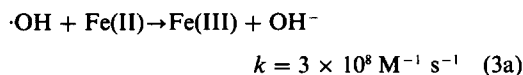
Reaction (1) has a quantum yield of ~ 1.0 per $\cdot\text{OH}$ radical formed (Hunt and Taube, 1952; Weeks and Matheson, 1956), but H₂O₂ absorbs UV radiation significantly only at wavelengths below 300 nm, and even then the molar absorption coefficients are low ($< 200 \text{ M}^{-1} \text{ cm}^{-1}$). For contaminated waters that absorb strongly in this region, there is a need for an AOP that can generate $\cdot\text{OH}$ radicals with near UV (300–400 nm) or even visible light ($> 400 \text{ nm}$). Such a system would also have application using solar radiation as the light source.

There are several processes involving iron compounds and hydrogen peroxide that provide alternative ways of generating hydroxyl radicals (Safarzadeh-Amiri *et al.*, 1996).

The dark reaction of Fe(II) with H₂O₂ known as Fenton's reaction (Fenton, 1894; Walling, 1975; Barbeni *et al.*, 1987) has been known for over a century.

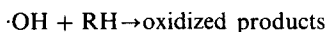


The hydroxyl radical thus formed can either react with Fe(II) producing Fe(III),



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†A few commercial processes employ the photolysis of ozone for this purpose.

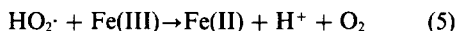
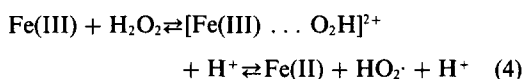
or react with and initiate the oxidation of organic pollutants present in solution:



$$k = 10^7 - 10^{10} \text{ M}^{-1} \text{ s}^{-1} \quad (3b)$$

Recent studies by Eisenhauer (1964), Barbeni *et al.* (1987), Sedlak and Andren (1991), Pignatello (1992) and Ruppert *et al.* (1993) have shown that a variety of organic pollutants, such as phenol, chlorophenols, chlorobenzene, etc., can be treated by Fenton's reagent. This process is only effective in an acidic pH range of ≤ 3.0 .

The decomposition of hydrogen peroxide is also catalyzed by ferric ions (Walling, 1975). In this process, H_2O_2 is decomposed to H_2O and O_2 , and a steady-state concentration of ferrous ions is maintained during peroxide decomposition, according to:



The Fe(II) ions react with H_2O_2 to generate hydroxyl radicals (reaction (2)), which then react with organic pollutants (reaction (3b)). The utility of the $\text{Fe(III)}/\text{H}_2\text{O}_2$ reagent to destroy organic pollutants in water media has also been demonstrated (Ruppert *et al.*, 1993; Sun and Pignatello, 1993; Murphy *et al.*, 1989; Bishop *et al.*, 1968). However, the initial rate of destruction of organic pollutants by the $\text{Fe(III)}/\text{H}_2\text{O}_2$ reagent has been found to be much slower than that for the $\text{Fe(II)}/\text{H}_2\text{O}_2$ reagent, perhaps due to the lower reactivity of ferric ion toward hydrogen peroxide. Again, this process is only effective at an acidic pH of about 2.8 (Pignatello, 1992).

Although both the $\text{Fe(II)}/\text{H}_2\text{O}_2$ and $\text{Fe(III)}/\text{H}_2\text{O}_2$ reagents are able to destroy organic pollutants in the dark, neither one is able to mineralize organic pollutants completely (Ruppert *et al.*, 1993; Bishop *et al.*, 1968; Murphy *et al.*, 1989; Pignatello, 1992). The starting material is partly transformed to some intermediate products, which appear to resist further oxidation reactions. This may be due in part to complexation of the intermediates formed with Fe(III) ions and in part to other competitive reaction routes with $\cdot\text{OH}$ radicals. The extent of mineralization was reported to be independent of the oxidation state of iron employed and about 40–60% mineralization is achieved with these reagents, depending on the amount of hydrogen peroxide added to the reaction mixture (Ruppert *et al.*, 1993; Pignatello, 1992).

The rate of destruction of organic pollutants and the extent of mineralization with the $\text{Fe(II)}/\text{Fe(III)}/\text{H}_2\text{O}_2$ reagent is improved considerably by irradiation with near-UV and visible light (Ruppert *et al.*, 1993; Pignatello, 1992; Cater *et al.*, 1992; Sedlak *et al.*, 1987; Haag and Yao, 1992; Lipczynska-

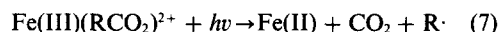
Kochany, 1992). Indeed, complete mineralization can be achieved in this manner. This process is called the "photo-Fenton reaction". The photo-enhancement is thought to arise from:

1. *Photoreduction of ferric ion to ferrous ion:* Irradiation of the hydroxylated ferric ion in aqueous solution produces ferrous ion and hydroxyl radicals (Balzani and Carassiti, 1970; Langford and Carey, 1975; Faust and Hoigné, 1990).



This is a wavelength dependent reaction and the quantum yield of formation of hydroxyl radical/ferrous ion decreases with increasing wavelength. For example, the quantum yield of $\cdot\text{OH}$ formation is 0.14 at 313 nm and 0.017 at 360 nm (Faust and Hoigné, 1990). In addition to hydroxyl radicals produced in reaction (6), the photogenerated Fe(II) can participate in the Fenton's reaction (reaction (2)), generating additional hydroxyl radicals, thus accelerating the rate of destruction of organic pollutants.

2. *Photodecarboxylation of ferric carboxylate complexes:* Fe(III) ions form stable complexes and/or associated ion pairs with carboxylates and polycarboxylates. These complexes are photochemically active and generate ferrous ions on irradiation according to (Balzani and Carassiti, 1970):

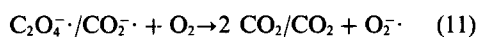
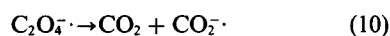
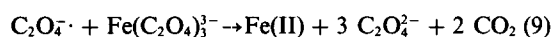
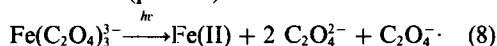


The radical $\text{R}\cdot$ can react with oxygen and degrade further. The ferrous ions can in turn react with added hydrogen peroxide (reaction (2)), generating hydroxyl radicals. Carboxylates are formed during the photocatalyzed oxidation of organic pollutants, thus photodecarboxylation, as shown in reaction (7), is expected to play an important role in the treatment and mineralization of organic molecules.

3. *The photolysis of hydrogen peroxide:* Some direct photolysis of H_2O_2 still occurs (reaction (1)); however, in the presence of strongly absorbing iron complexes, this reaction contributes only in a minor way to the photodegradation of organic contaminants.

The occurrence of photodecarboxylation reactions has been demonstrated (Pignatello, 1992; Sun and Pignatello, 1993; Balzani and Carassiti, 1970; Trott *et al.*, 1972; Frahn, 1958; Miles and Brezonik, 1981; Zauo and Hoigné, 1992; Faust and Zepp, 1993; Pehkonen *et al.*, 1993) and their significance in the aquatic environment has long been recognized. They are believed to be the cause of the under-saturation of oxygen in colored lake waters (Miles and Brezonik, 1981). Furthermore, this reaction is also believed to be responsible for the formation of hydrogen peroxide in cloud and rain waters (and in surface waters) and the destruction of organic molecules in these environments (Faust and Hoigné, 1990; Zauo and Hoigné, 1992; Faust and Zepp, 1993; Pehkonen *et al.*, 1993).

The quantum yield of Fe(II) formation in reaction (7) varies with different carboxylate ligands (Balzani and Carassiti, 1970). The ferrioxalate complex $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ is the best known and most widely studied example of Fe(III)-polycarboxylate complexes. It is a popular actinometer and is used widely to measure light intensity (Hatchard and Parker, 1956). Irradiation of ferrioxalate proceeds as follows in acid solution (pH $\sim$ 3):



The theoretical quantum yield for Fe(II) generation is 2.0, but reaction (11) and the reverse of reaction (8) reduce the actual quantum yield. The measured quantum yield of Fe(II) formation is about $\sim$ 1.0–1.2 from 250–450 nm and decreases with further increase of the irradiation wavelength (Balzani and Carassiti, 1970). In air saturated solution at pH 3, the intermediate oxalate radical reacts with molecular oxygen and generates the hydroperoxyl radical, which disproportionates to form hydrogen peroxide (Miles and Brezonik, 1981; Zauo and Hoigné, 1992; Faust and Zepp, 1993). The ferrous ion formed in reactions (8) and (9) can react with H_2O_2 present in solution and generate hydroxyl radicals (reaction (2)). Thus the photolysis of ferrioxalate in the presence of hydrogen peroxide provides a continuous source of Fenton's reagent.

Zepp *et al.* (1992) have shown that the photolysis of ferrioxalate in the presence of hydrogen peroxide generates hydroxyl radicals which can react with and oxidize organic pollutants present in solution. In a recent patent Safarzadeh-Amiri (1993) showed that irradiation of a ferrioxalate/ H_2O_2 mixture with UV-visible light is a very effective system for the destruction of various organic pollutants in water media.

Many wastewaters, which have moderate to high levels (several hundred up to a thousand mg/L) of pollutants, exhibit a high absorbance below 300 nm. This means that treatment with UV/ H_2O_2 will be ineffective due to the poor absorption of the UV photons by H_2O_2 . In these cases, the UV-vis/ferrioxalate/ H_2O_2 process provides advantages, since ferrioxalate has a high molar absorption coefficient for

wavelengths above 200 nm (see Fig. 1), absorbs light strongly at longer wavelengths (up to 500 nm) and generates hydroxyl radicals with a high quantum yield. Other carboxylate and polycarboxylate complexes of Fe(III) may not have spectral characteristics similar to those of ferrioxalate, and generally have much lower and wavelength dependent quantum yields as compared to ferrioxalate (Balzani and Carassiti, 1970).

The effective treatment of a waste stream does not necessarily require total mineralization of the organic pollutant(s), provided that no toxic intermediates are formed. Rather, the disappearance and partial oxidation of the pollutant(s) may be sufficient. Previous investigations have shown that a combination of UV light and hydrogen peroxide provides a powerful means of destroying and/or partially oxidizing organic pollutants in water media. The efficiency of oxidation in this process is a function of pollutant concentration and decreases with increasing concentration of pollutants and UV absorbance of the contaminated water. It can require a high concentration of hydrogen peroxide (Sun and Pignatello, 1993; Sundstrom *et al.*, 1992) and, for heavily contaminated and/or high UV light absorbing waters (absorbance $>$ 1.0 in 1 cm for $\lambda <$ 300 nm*), excessively large amounts of hydrogen peroxide ($>$ 1000 ppm) and large UV doses are required. An alternative treatment process that uses a lower UV dose and lower levels of hydrogen peroxide would offer significant advantages.

The UV-vis/Fe(II)/ H_2O_2 (UV-Fenton) process has been shown to be more efficient than the UV/ H_2O_2 process; however, it is only useful and economical for the treatment of waste streams contaminated with low to moderate concentrations of pollutants with low UV absorbance. This is again due to: (a) the fraction of light absorbed by Fe(II) in the presence of a high concentration of pollutant and/or in highly absorbing water is small, requiring long irradiation times, and (b) scavenging of $\cdot\text{OH}$ radicals by high

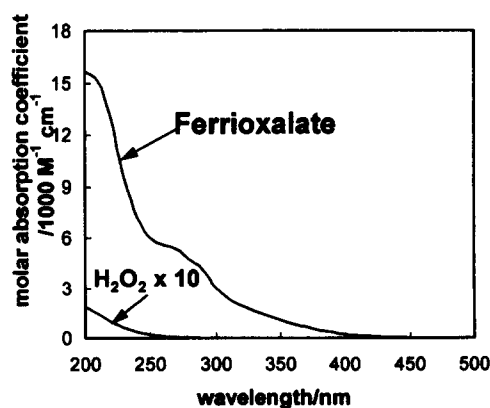


Fig. 1. Absorption spectra of hydrogen peroxide and potassium ferrioxalate in aqueous solution.

*For waters that are contaminated with pollutants that absorb strongly in the near UV and visible wavelength regions (e.g. textile dyes), no photochemical process would be very efficient because of competition for photons.

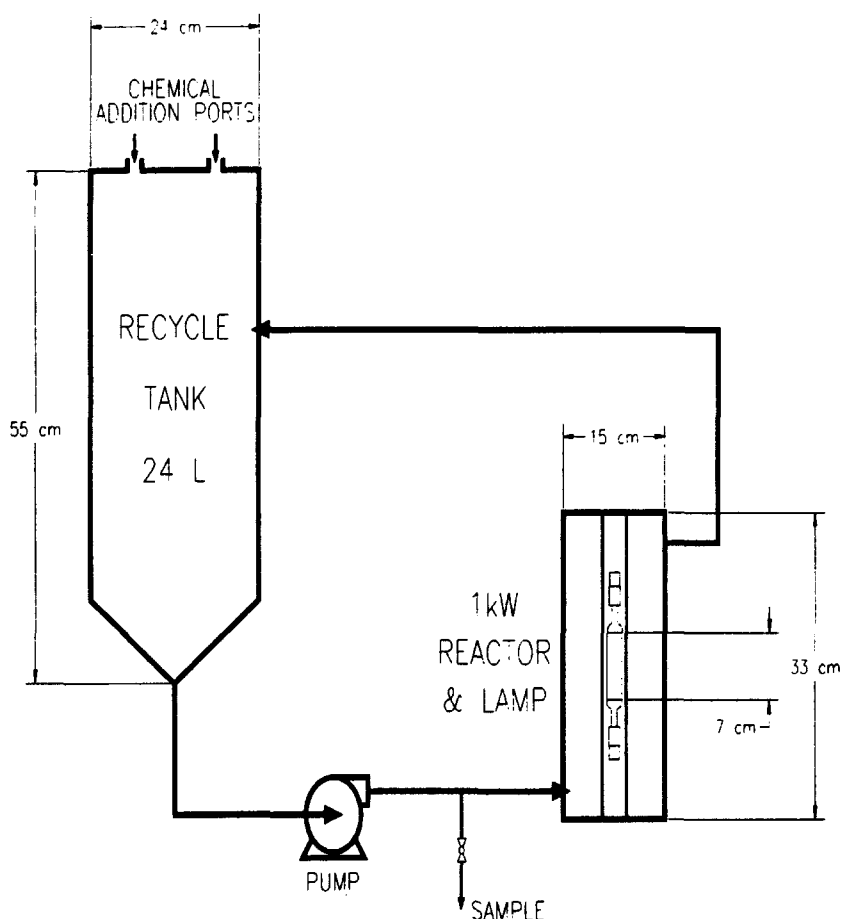


Fig. 2. Schematic of the UV reactor and supplementary components comprising the AOP unit.

concentrations of hydrogen peroxide, dissolved non-organic materials and any other $\cdot\text{OH}$ scavengers.

In this paper, we present the results of our investigations on the applications of ferrioxalate as a photo-mediator (the UV-vis/ferrioxalate/ H_2O_2 process) for the treatment of industrial wastewater and contaminated groundwater. In particular, we make direct comparisons of treatment efficiencies among the UV/ H_2O_2 , the UV-vis/ $\text{Fe(II)}/\text{H}_2\text{O}_2$ and the UV-vis/ferrioxalate/ H_2O_2 processes.

EXPERIMENTAL AND METHODS

Materials

Technical grade ferric and ferrous sulfate and oxalic acid were obtained from A&K Petrochem (Toronto, Ontario, Canada) and were used without further purification. Other chemicals were reagent grade (from Aldrich) and were used as received.

General procedure

The contaminated water was treated using a bench scale AOP reactor without pretreatment unless otherwise indicated. The same reactor with the same UV lamp was employed in all experiments described here. The AOP reactor operates under batch recirculation mode, where the water is added to a holding tank and then recirculated

through a 16 cm diameter cylindrical reactor housing the UV lamp enclosed in a 4.2 cm diameter quartz sleeve. The reactor volume is 5.8 L and the total system volume (reactor plus holding tank and piping) is 30 L. Wetted parts are 316 L stainless steel, viton and quartz. A schematic of the AOP unit is shown in Fig. 2. In a typical run, the test unit was loaded with about 30 L of contaminated water and then recirculated through the reactor at 115 L/min. The solution was spiked with additional pollutants (see Tables 3–5), if necessary, and 5–30 min (depending on the solubility of organic pollutants) was allowed for the dissolution of the spiked pollutants before an initial sample was taken prior to the addition of any reagent. The pH was adjusted using sulfuric acid and/or sodium hydroxide, then the ferrioxalate/ H_2O_2 (or other reagents) were added and the lamp turned on. Samples were taken at regular time intervals for the determination of the concentration of various contaminants as a function of light dose and were analyzed immediately. The reaction was quenched (for the UV/Fenton and UV-vis/ferrioxalate/ H_2O_2 processes) with either addition of NaOH to raise the pH to precipitate ferric ions or sodium sulfite to destroy the hydrogen peroxide.

Photolysis

The light source was a 1 kW medium pressure Hg UV lamp specifically manufactured for Calgon Carbon Oxidation Technologies (CCOT), which emits in the range 200–600 nm. Approximately 29% of the emitted light power is in the range 200–300 nm with a further 25% between 300 and 500 nm (see Table 1).

Table 1. The output of the 1000 W CCOT lamp in 200–500 nm region

Wavelength region (nm)	Light output (W)
200–249	173
250–299	117
300–349	42
350–399	72
400–449	53
450–500	66

Analysis

The concentrations of benzene, toluene and xylenes (BTX), chlorobenzene, PCE and TCE were determined by high performance liquid chromatography (HPLC), consisting of a Waters-Associates Model 6000 A solvent delivery system pump, a variable wavelength Waters Lambda-Max Model 481 LC absorbance detector and a Supelco ODSII Spherisorb S50DS2 C₁₈ reverse phase column (15 cm long, 0.46 cm i.d.). The mobile phase was a mixture of acetonitrile (60%) and water (40%) at a flow rate of 1.5 mL/min. The other pollutants (methanol, 2-butanone and 1,4-dioxane) were measured by gas chromatography using a Varian Model 3300 GC equipped with a flame ionization detector (FID) and a J&W Scientific DB-WAX, DB-5 or DB-624 megabore column (30 m long and 0.53 mm i.d.). Helium was used as carrier gas. Additionally, analyses of 1,4-dioxane were confirmed by an outside commercial laboratory using EPA method 624. Mineralization was monitored by the reduction of the total organic carbon (TOC) in solution and the biological oxygen demand (BOD*) was measured as BOD₅, in which the contaminated water, or an appropriate dilution, was incubated for 5 days at 20°C in the dark. Both analyses were carried out by an outside commercial laboratory. The BOD₅ value was calculated for samples that contained approximately 35–75% of the original oxygen content. Oxygen concentration was measured with an oxygen electrode using a YSI DO meter, Model 50B and YSI DO probe, Model 5730. The electrode tip was covered with Teflon, 0.5 mm thickness. Literature values (Verschueren, 1983) for the contribution to BOD by various pollutants were used to estimate the total solution BODs. For TOC measurements, the sample was introduced into an auto-analyzer system, where it was acidified and flushed with nitrogen at 500 mL/min to remove inorganic carbon. The sample was then oxidized with a UV digester in an acid persulfate medium. The carbon was converted to carbon dioxide and following dialysis, was measured by determining the loss of absorbance of a weakly buffered alkaline phenolphthalein solution using an AAI colorimeter operating in the inverse mode. Formic acid was determined as formate by ion chromatography using a Waters IC-Pak Ion-exclusion column (7.8 × 300 mm) and a Waters Model 431 conductivity detector. The mobile phase was a mixture of sodium borate/gluconate concentrate (0.02%), *n*-butanol (0.02%), acetonitrile (0.12%) and water (84%) at a flow rate of 1 mL/min. The sodium borate/gluconate concentrate was prepared by dissolving 16 g sodium borate, 18 g boric acid and 25 g sodium tetraborate decahydrate in approximately 500 mL of deionized water. To this solution was added 250 mL of glycerin, and the volume was brought to 1 L with deionized water.

Energy efficiency figure-of-merit

When comparing treatment efficiencies for various processes, it is important to use a figure-of-merit that will

*Biological oxygen demand (BOD) is the amount of oxygen that is required by microorganisms for the oxidation of organic chemicals present in wastewater and varies for different compounds.

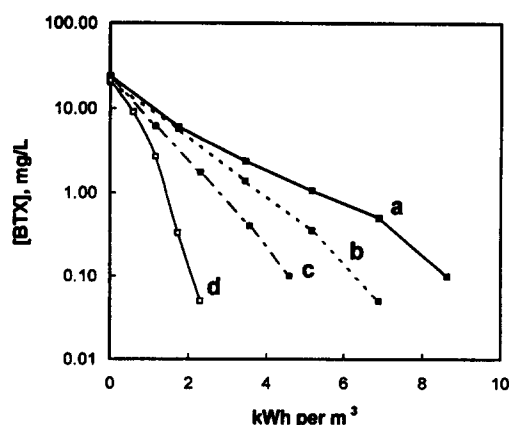


Fig. 3. Semilog plot of the destruction of total BTX concentration with the UV/H₂O₂ process—(a) [H₂O₂] = 50 mg/L, (b) [H₂O₂] = 300 mg/L, (c) [H₂O₂] = 600 mg/L—and the UV-vis/ferrioxalate/H₂O₂ process—(d) [H₂O₂] = 50 mg/L and [Fe(III)] = 5 mg/L.

be unbiased and also directly related to treatment costs. For this purpose we have used the *Electrical Energy per Order* (Bolton and Cater, 1994; Bolton *et al.*, 1996), defined as the electrical energy in kilowatt hours (kWh) required to degrade a contaminant by one order of magnitude in 1 cubic m of contaminated water. It can be calculated (for a batch treatment) using the following formula:

$$EE/O = \frac{P \times t \times 1000}{V \times \log(c_i/c_f)} \quad (12a)$$

where P is the lamp power (kW) of the unit, V is the volume (L) of water treated in time t (h) and c_i and c_f are the initial and the final concentrations of pollutant. The lower the EE/O value, the more efficient is the treatment process. The EE/O concept implicitly assumes first-order kinetics. Thus, if the first-order rate constant is k_1 (min⁻¹) (Bolton and Cater, 1994), then the EE/O is given by:

$$EE/O = \frac{38.4P}{V \times k_1} \quad (12b)$$

When the reaction kinetics deviate from first-order behavior, as is the case for most of the examples in this paper, an average EE/O may be obtained from the initial and final concentrations. EE/Os obtained in this manner may be used for comparative purposes but cannot be extrapolated to other concentration levels.

RESULTS AND DISCUSSION

Spiked tap water tests

Since the UV-vis/ferrioxalate/H₂O₂ process generates hydroxyl radicals (reactions (8), (9) and (2)), any organic pollutant that is degraded in the UV/H₂O₂ process (i.e. degrades through reaction with hydroxyl radicals) should also function with the UV-vis/ferrioxalate/H₂O₂ process. However, there is a question of process efficiency. To examine this question, we have carried out tests on tap water spiked with BTX (benzene, toluene and xylenes). Figure 3 shows the destruction of total BTX with the UV/H₂O₂ process at various concentrations of hydrogen peroxide. For comparison, this figure also shows the destruction of BTX with the UV-vis/ferrioxalate/H₂O₂ process. The treatment efficiency of the UV/H₂O₂ process increases

Table 2. Effect of H_2O_2 concentration on the UV/ H_2O_2 treatment efficiency of BTX^a in spiked tap water

[H_2O_2] (mg/L)	EE/O ^b (kWh/order/m ³)	% Light absorbed by H_2O_2
50	4.0	18
100	3.7	32
200	3.2	47
300	2.9	57
600	2.0	75

^aThe average concentration of each individual pollutant was about 7 mg/L and the initial solution pH was adjusted to about 3.5.

^bAverage EE/O for the destruction of benzene, toluene and xylenes.

with increasing concentration of hydrogen peroxide, due to absorption of a greater fraction of UV light by hydrogen peroxide in untreated water. However, the improvement of the treatment efficiency is not proportional to the fraction of light absorbed by H_2O_2 in untreated water. The data presented in Table 2 indicate that increasing the fraction of light absorbed by hydrogen peroxide by a factor of 4 results in a reduction of the EE/O by only a factor of 2. This may be attributed to the following:

(a) The fraction of light absorbed by hydrogen peroxide can decrease dramatically during the treatment process, due to the formation of phenolic intermediates, which absorb more strongly than the parent compounds in the 200–300 nm region. This feature is illustrated in Fig. 4 where the absorption spectrum of a solution containing ~30 mg/L of BTX is shown before and after 9 min of irradiation in the presence of 50 mg/L of H_2O_2 .

(b) Scavenging of hydroxyl radicals by H_2O_2 . This scavenging becomes more significant as the concentration of hydrogen peroxide increases. For example, increasing the concentration of hydrogen peroxide from 50 to 600 mg/L reduces the ratio of the rates of reaction of hydroxyl radical with benzene versus that with hydrogen peroxide from 18 to about 1.5, respectively.

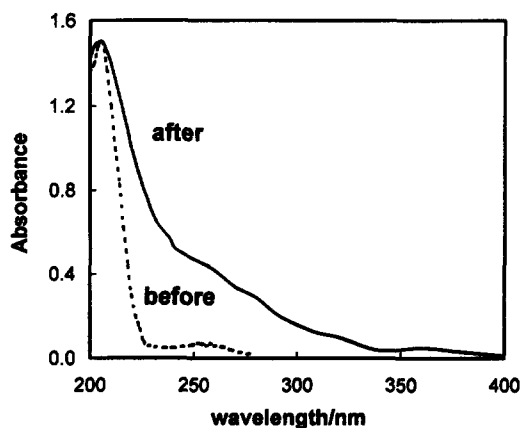


Fig. 4. Absorption spectra of a solution containing about 30 mg/L of total BTX before and after 9 min of irradiation with UV-vis light in the presence of 50 mg/L of H_2O_2 .

We have carried out additional tests on spiked tap water for several pollutants and have compared the treatment efficiencies for the UV/ H_2O_2 , UV-vis/ $\text{Fe(II)/H}_2\text{O}_2$ and UV-vis/ferrioxalate/ H_2O_2 processes. The results are summarized in Table 3. These experiments were carried out under identical experimental conditions (same apparatus, UV lamp and temperature). In all cases, the process conditions have been chosen so as to achieve near optimum results in terms of operational cost and equipment size. In our experience, optimum results (from an economic standpoint) are achieved when about 40% of incident UV-vis radiation is absorbed by the active reagent(s) and the reaction is carried out at an acidic pH, particularly for the UV-vis/ $\text{Fe(II)/H}_2\text{O}_2$ and UV-vis/ferrioxalate/ H_2O_2 processes. The data presented in Table 3 indicate that the latter process is more efficient (by a factor of 2–15) than the other two for treating BTX, 1,4-dioxane and methyl-*t*-butyl ether (MTBE). However, its advantage for the destruction of chloroalkenes tested in this investigation appears to be less significant, as far as the spiked tap water conditions are concerned. The efficiency of the UV-vis/ferrioxalate/ H_2O_2 processes is marginally greater than the UV-vis/ $\text{Fe(II)/H}_2\text{O}_2$ process and by a factor of about two greater than that of the UV/ H_2O_2 process.

Treatment of industrial wastewater

Several representative wastewaters of varying degrees of contamination, background absorption, alkalinity, etc., have been treated with UV/ H_2O_2 , UV-vis/ $\text{Fe(II)/H}_2\text{O}_2$ and UV-vis/ferrioxalate/ H_2O_2 . The treatment efficiencies for these processes, along with water characteristics and the reagent concentrations, are presented in Tables 4 and 5. As can be seen, the UV-vis/ferrioxalate/ H_2O_2 process is more efficient by a factor of 2–7 than the other two processes.

In order to provide further details on the comparative treatment of wastewaters, we present five examples. It is important to note that these are difficult waters to treat, since in each case there are significant concentrations of background components.

Example 1: Figure 5 shows the destruction of chlorobenzene (CB) in a polluted process water (water sample No. 1) by the UV/ H_2O_2 , the UV-vis/ $\text{Fe(II)/H}_2\text{O}_2$ and the UV-vis/ferrioxalate/ H_2O_2 processes. The water is clear with a low UV absorbance, an alkalinity of ~400 mg/L, a COD of ~150 mg/L with ~570 mg/L of chloride ion and 150 mg/L chlorobenzene. Although conditions for the first two processes have been chosen for near optimum results, these treatments are not as efficient (EE/Os of 9.6 and 3.0 kWh/order/m³, respectively) for the destruction of CB, compared with the UV-vis/ferrioxalate/ H_2O_2 process (EE/O of 1.14 kWh/order/m³).

Example 2: Figure 6 shows the destruction of benzene, toluene and xylenes in a settled tank bottom

Table 3. Comparison of treatment efficiencies for spiked tap water

Pollutant	Concentration (mg/L)	EE/O (kWh/order/m ³) ^a		
		UV/H ₂ O ₂ ^b	UV-vis/ Fe(II)/H ₂ O ₂ ^c	UV-vis/ Ferrioxalate/H ₂ O ₂ ^d
BTX	~7 each ^e	1.8	1.2	0.6
	~80 each ^f	20.0	2.4	1.3
1,4-Dioxane ^g	100	3.2	2.3	0.7
MTBE ^h	100	6.3	2.9	1.8
TCE ⁱ	100	2.1	1.3	0.9
PCE ^j	100	3.7	2.7	1.8

^aThe EE/O values are accurate to within 10% and are average values, since they have been calculated from initial and final concentrations. They have been corrected the same as in Table 5.

^bInitial pH = 3.5 and did not change significantly during the run.

^cInitial pH = 3.0 and did not change significantly during the run.

^dInitial pH = 2.8, pH increases as treatment progresses and was kept around 2.8–3.2.

^e[H₂O₂] = 600 mg/L for the UV/H₂O₂ process; for other two processes [H₂O₂] = 50 mg/L, [Fe(II)] = 10 mg/L, [Fe(III)] = 10 mg/L.

^f[H₂O₂] = 2000 mg/L for the UV/H₂O₂ process; for other two processes [H₂O₂] = 500 mg/L, [Fe(II)] = 10 mg/L, [Fe(III)] = 10 mg/L.

^g[H₂O₂] = 250 mg/L for the UV/H₂O₂ and UV-vis/Fe(II)/H₂O₂ processes, and 200 mg/L for the UV-vis/ferrioxalate/H₂O₂ process, [Fe(II)] = 40 mg/L, [Fe(III)] = 40 mg/L.

^hMethyl-*t*-butylether (MTBE), [H₂O₂] = 500 mg/L, [Fe(II)] = 10 mg/L, [Fe(III)] = 10 mg/L.

ⁱThe tap water was spiked with both trichloroethylene (TCE) and perchloroethylene (PCE) and were treated together. [H₂O₂] = 500 mg/L, [Fe(II)] = 40 mg/L, [Fe(III)] = 40 mg/L.

water (sample No. 2). The polluted water has a COD of ~1000 mg/L, absorbs significantly in the UV and near-UV region (for a pathlength of 1.0 cm, the absorbance is 0.48 at 440 nm, 1.0 at 350 nm and >2 below 280 nm, where the water absorbs completely) and has significant amounts of suspended solids. For these reasons, this contaminated water could not be treated efficiently with either the UV/H₂O₂ or UV-vis/Fe(II)/H₂O₂ processes. However, as can be seen in Fig. 6, the UV-vis/ferrioxalate/H₂O₂ process is able to treat this wastewater with an EE/O of about 1.6 kWh/order/m³.

Example 3: A further example of enhancement of treatment efficiency was provided by wastewater (sample No. 3) contaminated with ~500 mg/L of 1,4-dioxane, soap and other pollutants and had a COD of ~1500 mg/L. It was cloudy, white and was partly opaque to UV-visible light. The treatment objective was the reduction of 1,4-dioxane concentration to below 0.5 mg/L and that of TOC to less than 50 mg/L. Figure 7 shows the destruction of 1,4-dioxane versus UV dose by the UV/H₂O₂,

UV-vis/Fe(II)/H₂O₂ and UV-vis/ferrioxalate/H₂O₂ processes. Note that because of the high starting concentration, the kinetics tend to be zero-order at the start and change over to first-order after some time. The EE/Os of the first two processes are similar (EE/O ≈ 7.4 kWh/order/m³) and more than twice that of the UV-vis/ferrioxalate/H₂O₂ process (EE/O = 3.2 kWh/order/m³).

Figure 8 compares the destruction of TOC by the UV/H₂O₂ and UV-vis/ferrioxalate/H₂O₂ processes. Although 1,4-dioxane was destroyed by both processes (see Fig. 7), the TOC concentration remained constant in the former process over the duration of the experiment. The UV-vis/ferrioxalate/H₂O₂ process, on the other hand, was able to bring about a substantial mineralization of the pollutants present in this water (84% of the original organic carbon was mineralized). Oxidation by the UV/H₂O₂ process did not lead to any mineralization of the 1,4-dioxane and other pollutants present in this water (over the time scale of experiment), as can be seen from the TOC data (curve a in Fig. 8). The pollutants,

Table 4. The characteristics of various polluted waters treated in this study

Water type	Water sample no.	Compound(s) ^a	Initial conc. (mg/L)	COD (mg/L)	Absorbance ^b (λ < 300 nm)	[H ₂ O ₂] (mg/L)	[Fe ³⁺] ^c (mg/L)
Process water	1	Chlorobenzene	150	150	Low	300	20
Tank bottom water	2	BTX ^d	10	520	High	500	60
			(each)				
Process water	3	1,4-Dioxane	450	1480	Moderate	3000	40
Groundwater	4	DCE ^e	30	290	Moderate-high	800	40*
Groundwater	4	MEK ^f	18	290	Moderate-high	400	40*
Process water	5	Methanol/formaldehyde	250	450	Low	2000	40

^aThere were other pollutant(s) beside the ones indicated in all of these polluted waters.

^bLow, moderate and high absorbance means $A < 0.5$, $0.5-1.5$, and > 1.5 , respectively.

^cThe concentration of oxalic acid was always three times the molar concentration of ferric ion.

^dBenzene, toluene, xylenes.

^e1,2-Dichloroethylene.

^fMethylethylketone.

*Present as dissolved iron in this water sample.

Table 5. The EE/O values^a (kWh/order/m³) for the destruction of pollutants listed in Table 4 by the various AOPs

Primary pollutant	Water sample no.	UV/H ₂ O ₂	UV-vis/ Fe(II)/H ₂ O ₂	UV-vis/ ferrioxalate/H ₂ O ₂
Chlorobenzene	1	9.6	3.0	1.4
BTX	2	6.6	3.7	1.6
1,4-Dioxane	3	7.4	7.4	3.1
DCE ^b	4	6.6	2.2	0.6
MEK ^b	4	12.4	11.9	4.2
Methanol	5	14.2	10.3	3.6

^aThe EE/O values in this table have been corrected using a sophisticated kinetic model that corrects for batch to flow errors caused by significant depletion of the contaminant per pass through the reactor, so that these EE/Os are appropriate for a low-through configuration. These EE/Os are average values (see footnote a in Table 3).

^bUsed the natural iron present in this water sample in UV-vis/Fe(II)/H₂O₂ and UV-vis/ferrioxalate/H₂O₂ processes.

therefore, were only oxidized to some intermediate products, which were resistant, on the time scale of our experiment, to further oxidation. Reduction of TOC with the UV/H₂O₂ process may be possible with further irradiation over a longer period of time and addition of higher concentrations of hydrogen peroxide, but the process would then not be economical. Note that in the UV-vis/ferrioxalate/H₂O₂ process, initial TOC is higher due to the addition of organic carbon in the form of oxalate in this process. The oxalate is mineralized rapidly through photodecarboxylation reaction (reactions (8)–(11)) and at the same time oxidizes 1,4-dioxane. Oxidation of the latter compound generates carboxylic acids which are resistant to further oxidation. However, these intermediates complex with Fe(III) and these complexes are photoactive and degrade further through decarboxylation (reactions (7) and (8)–(11)).

Example 4: Here we provide an example of a

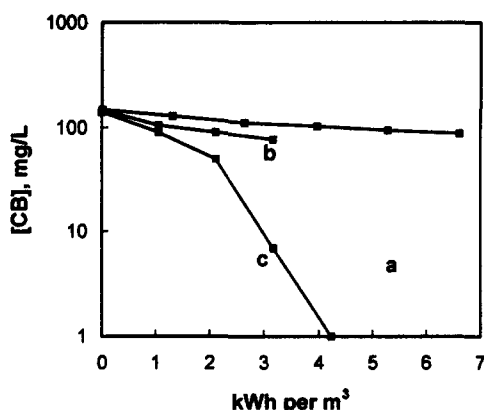


Fig. 5. Semilog plot of the treatment versus UV dose (kWh per m³) of chlorobenzene (CB) in a process water with (a) the UV/H₂O₂ process (pH = 6.5, [H₂O₂] = 500 mg/L), (b) the UV-vis/Fe(II)/H₂O₂ process (pH = 3, [H₂O₂] = 300 mg/L and [Fe(II)] = 20 mg/L), and (c) the UV-vis/ferrioxalate/H₂O₂ process (pH = 2.8, [H₂O₂] = 300 mg/L, [Fe(III)] = 20 mg/L, [oxalic acid] = 276 mg/L). The x axis represents the UV dose expressed as the consumption of electrical energy (kWh) per m³ of contaminated water.

contaminated groundwater that was treated with various processes. The water was contaminated with about 300 mg/L of various pollutants including about 30 mg/L of dichloroethylene and about 20 mg/L of 2-butanone. It has a low alkalinity (36 mg/L), has a moderate to high absorption in the UV region and contains about 40 mg/L of dissolved iron. The dissolved iron was precipitated by raising the pH to 7, and the iron free solution was then treated with the UV/H₂O₂. The data shown in Fig. 9 and presented in Table 5 indicate that this process is not efficient for treating these pollutants. The treatment efficiency is improved slightly (by a factor of about 3 for DCE and by about 17% for MEK) with the UV-vis/Fe(II)/H₂O₂ process. However, as the data presented in Table 5 (and shown in Fig. 9) show, the treatment efficiency is improved considerably with the UV-vis/ferrioxalate/H₂O₂ process. The EE/O for the oxidation of DCE and MEK in the latter process is reduced by more than an order of magnitude and a factor of 7, respectively, as compared to the UV/H₂O₂ process.

Example 5: In this example, the reduction of BOD of an industrial wastewater (sample No. 5) to below

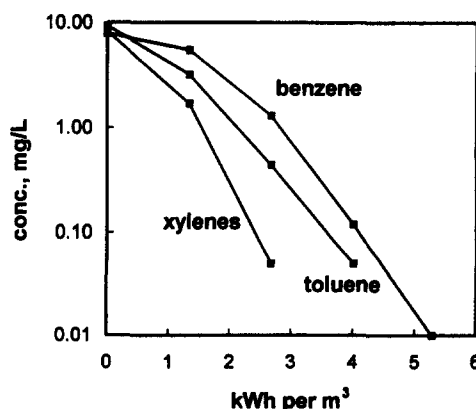


Fig. 6. Semilog plot of the treatment versus UV dose (kWh per m³) of benzene, toluene and xylenes in a tank bottom water by the UV-vis/ferrioxalate/H₂O₂ process at pH = 2.8 with [Fe(III)] = 60 mg/L, [oxalic acid] = 405 mg/L and [H₂O₂] = 500 mg/L.

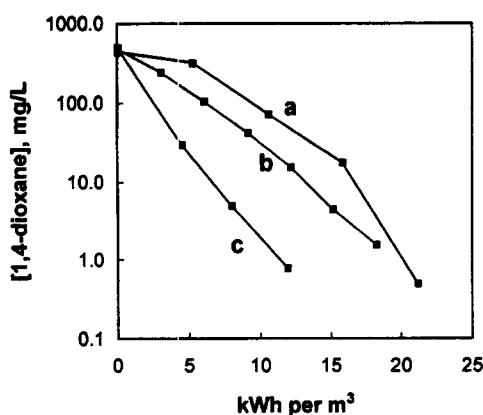


Fig. 7. Semilog plot of the destruction of 1,4-dioxane versus UV dose (kWh per m^3) in a process water with (a) the $\text{UV/H}_2\text{O}_2$ process ($\text{pH} = 7$, $[\text{H}_2\text{O}_2] = 3000 \text{ mg/L}$), (b) the $\text{UV-vis/Fe(II)/H}_2\text{O}_2$ process ($\text{pH} = 3.0$, $[\text{H}_2\text{O}_2] = 3000 \text{ mg/L}$, $[\text{Fe(II)}] = 30 \text{ mg/L}$) and (c) the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process ($\text{pH} = 2.8$, $[\text{Fe(III)}] = 40 \text{ mg/L}$, $[\text{oxalic acid}] = 276 \text{ mg/L}$, $[\text{H}_2\text{O}_2] = 2000 \text{ mg/L}$).

1 mg/L was required. The water was contaminated with $\sim 250 \text{ mg/L}$ each of methanol and formaldehyde and about 20–50 mg/L of formic acid with a low background COD and absorptivity. These are simple organic compounds and can be treated by a variety of techniques, including chemical oxidation (using $\text{H}_2\text{O}_2/\text{NaOH}$ (Junkermann and Schwab, 1978)), oxidation at high temperature and pressure ("Wetox"), biological treatment and by the Fenton's reagent (Murphy *et al.*, 1989). However, there are problems associated with these processes, including: (a) the first three are slow, (b) the oxidation could be incomplete with the chemical oxidation process leading to formation of formic acid or formate ions and (c) the Fenton and Fenton-like process (Sedlak and Andren, 1991; Ruppert *et al.*, 1993; Bishop *et al.*, 1968; Murphy *et al.*, 1989) require a high concentration of hydrogen peroxide and Fe(II)

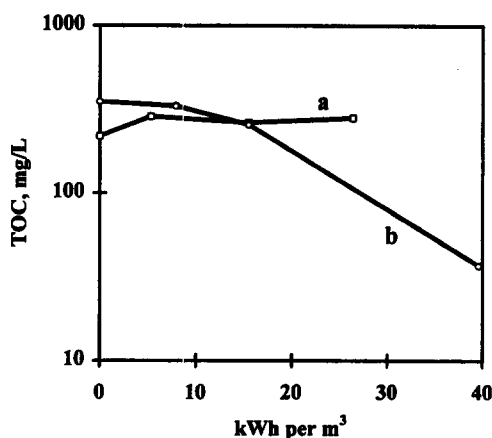


Fig. 8. Semilog plot of the reduction of total organic carbon (TOC) versus UV dose (kWh per m^3) with (a) the $\text{UV/H}_2\text{O}_2$ process and (b) the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process in the 1,4-dioxane contaminated water used for Fig. 6.

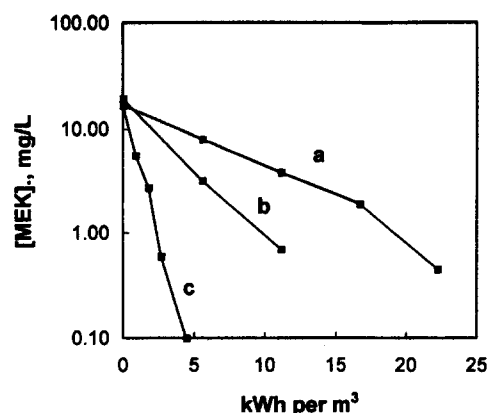


Fig. 9. Semilog plot of the destruction of 2-butanone (MEK) vs UV dose (kWh per m^3) in a contaminated ground water with (a) the $\text{UV/H}_2\text{O}_2$, (b) the $\text{UV-vis/Fe(II)/H}_2\text{O}_2$ and (c) the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ processes.

and/or Fe(III) ions. All of these pollutants can be treated easily and efficiently with AOP processes, as is illustrated in Fig. 10. Figure 11 shows the reduction of BOD and TOC by the $\text{UV/H}_2\text{O}_2$ and $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ processes.

As can be seen, both the $\text{UV/H}_2\text{O}_2$ and $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ processes are efficient in destroying these chemicals and in reducing the total BOD. However, the latter process is more than twice as efficient as the former in reducing the BOD. The TOC was also reduced to below 3 mg/L over the same treatment period. The efficiency of the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process for treating this water was initially similar to that of the $\text{UV/H}_2\text{O}_2$ process and the efficiency improved as the oxidation of methanol and formaldehyde continued.

The higher efficiency of the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process, as compared to the $\text{UV/H}_2\text{O}_2$ process,

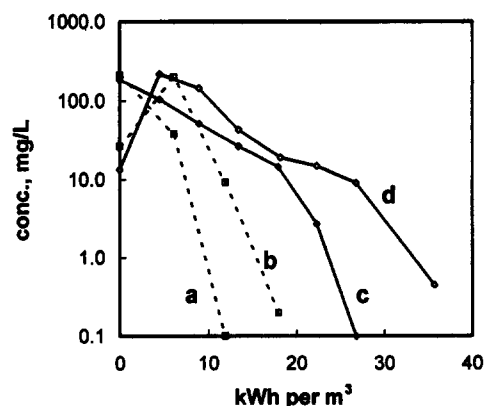


Fig. 10. Semilog plot dependence of the methanol (curves a and c) and formic acid (curves b and d) concentrations versus the UV dose (kWh per m^3) in a wastewater treated with the $\text{UV/H}_2\text{O}_2$ process ($\text{pH} = 4.0$, $[\text{H}_2\text{O}_2] = 1200 \text{ mg/L}$) (curves c and d) and the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process ($\text{pH} = 2.8$, $[\text{H}_2\text{O}_2] = 2000 \text{ mg/L}$, $[\text{Fe(III)}] = 40 \text{ mg/L}$, $[\text{oxalic acid}] = 756 \text{ mg/L}$) (curves a and b).

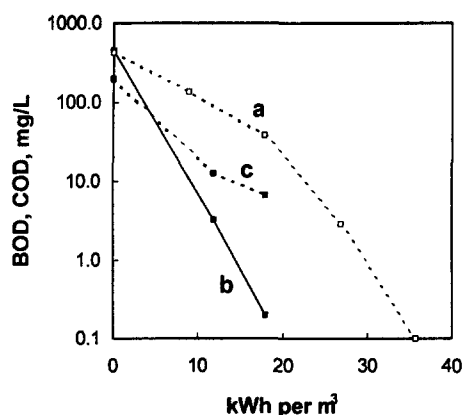


Fig. 11. Semilog plot of the reduction of BOD versus UV dose (kWh per m^3) with the $\text{UV/H}_2\text{O}_2$ process (curve a) and the reduction of BOD (curve b) and TOC (curve c) vs UV dose (kWh per m^3) with the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process in the wastewater used for Fig. 8.

is not surprising, considering the fact that (a) the rate of production of hydroxyl radicals by irradiation of the ferrioxalate/ H_2O_2 mixture is higher than that by UV irradiation of hydrogen peroxide and (b) the oxidation of methanol and formaldehyde generates formic acid, which can react and form complexes with Fe(III) . This complex is photoactive (Balzani and Carassiti, 1970) and decomposes according to reaction (7), generating Fe(II) . The Fe(II) reacts with hydrogen peroxide to generate hydroxyl radicals (reaction (2)), which then react with and oxidize methanol and formaldehyde to formic acid. The formation and destruction of formic acid are shown in Fig. 11.

Why is the UV-vis/ferrioxalate/ H_2O_2 process more efficient than the other processes?

The examples presented in this paper (see Tables 2–5 for treatment of spiked tap water and groundwater contaminated with various pollutants) show that the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process is much more efficient for destroying organic pollutants than are the $\text{UV-vis/Fe(II)/H}_2\text{O}_2$ or the $\text{UV/H}_2\text{O}_2$ processes. The reasons for the improved performance of $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process are believed to be as follows:

1. The quantum yield of Fe(II) formation is ~ 1.0 – 1.2 over a broad range of irradiation wavelength (i.e. from 250 to 450 nm) (Balzani and Carassiti, 1970). This means that at least one $\cdot\text{OH}$ radical can be formed for every quantum of light energy absorbed by ferrioxalate. Indeed, Zepp *et al.* (1992) have recently demonstrated that the ratio of photoformed hydroxyl radicals to ferrous ion is about unity*.

*Preliminary measurements indicate that under certain circumstances the quantum yield for the generation of $\cdot\text{OH}$ radicals can be much greater than unity in the $\text{UV-vis/ferrioxalate/H}_2\text{O}_2$ process. This work will be published separately.

2. Ferrioxalate absorbs over a broad range of wavelengths (see Fig. 2), and thus utilizes the UV-visible output of the lamp more efficiently.

3. Photolysis of ferrioxalate produces Fe(II)-oxalate species, as well as ferrous ion, and these species react very fast with H_2O_2 (Rush *et al.*, 1990) and generate hydroxyl radicals.

4. The process is more efficient for waters contaminated with aromatic pollutants because hydroxy and poly-hydroxy aromatic compounds (i.e. phenols, hydroquinone, etc., from benzene and chlorobenzene) are formed in the initial stages of oxidation of aromatic molecules. These intermediate oxidation products absorb strongly (see Fig. 3) in the UV and near-UV region and thus compete with hydrogen peroxide and Fe(III) for UV and near-UV light, leading to a very slow rate of destruction of the parent molecule. Ferrioxalate, on the other hand, absorbs strongly from 250 nm to 500 nm, thus blocking of the UV and near-UV portion of spectrum by the oxidation intermediates still leaves the visible portion of light energy to be absorbed by ferrioxalate.

5. The oxidation of organic compounds is known to generate aliphatics. For example, $\text{UV/H}_2\text{O}_2$ oxidation of phenol produces fumaric, formic, maleic, malic and oxalic acids (Scheck and Frimmel, 1995). Similar intermediates are expected to be formed in ferrioxalate mediated process too. Table 6 provides the rate constants for reactions of hydroxyl radicals with organic acids and the photoreduction quantum yield of their ferric ion complexes. While the rate of reaction of fumaric, formic and maleic acids with hydroxyl radical is very high, those of malic acid and oxalic acid are moderate to very slow. All of these compounds absorb in the UV region, reducing the fraction of light absorbed by hydrogen peroxide, thus reducing the yield of hydroxyl radical formation. On the other hand, these and other reaction intermediates can complex with Fe(III) (Pignatello, 1992; Sun and Pignatello, 1993), which are absorbing strongly in the near-UV and visible region. Some of these complexes can be photolyzed and produce ferrous ions with moderate to high quantum yield and start a new round of Fenton reaction.

Table 6. Rate constant for reactions of hydroxyl radicals with organic acids and the photoreduction quantum yield of Fe(III)-organic complexes

Organic acid	k ($\text{L mol}^{-1} \text{s}^{-1}$)	$\phi_{\text{Fe(II)}}$
Formic acid	9.6×10^9	0.05–0.55 ^a
Fumaric acid	6×10^9	—
Maleic acid	6×10^9	0.2–0.29 ^b
Malic acid	7.3×10^8	—
Oxalic acid	1.4×10^6	1.0–1.2

^aQuantum yield varies with concentration of formic acid and irradiation wavelength (Baxendale and Bridge, 1955).

^bQuantum yield varies with pH (Abrahamson *et al.*, 1994).

It is for these reasons that the UV-vis/ferrioxalate/ H_2O_2 process gives much lower EE/O values compared to other treatment processes.

CONCLUSIONS

A new photodegradation process (UV-vis/ferrioxalate/ H_2O_2), mediated by a mixture of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ and H_2O_2 , is introduced for the treatment of contaminated water streams. A variety of organic pollutants can be totally degraded by this process. The main theme of this study was to demonstrate the usefulness and the range of applicability of the UV-vis/ferrioxalate/ H_2O_2 process, as compared to the UV/ H_2O_2 and UV-vis/ $\text{Fe}(\text{II})/\text{H}_2\text{O}_2$ processes, for the treatment of moderate to heavily contaminated water streams. We have shown that for the cases presented, this new process is more efficient (by a factor of 2.5 to 30 times), based on energy use, as compared to other photodegradation processes examined here.

Since the ferrioxalate mediated process requires less electrical energy than the UV/ H_2O_2 process, this should generally result in lower treatment costs, since fewer lamps would be required. There may be some cases, however (e.g. low UV absorbing waters, low pollutant concentrations), where the UV/ H_2O_2 process would still offer the more cost effective solution, due to lower system complexity and reagent costs.

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